

On Parameters Influencing Voters' Preference Dynamics In The Indian Context

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Abstract

In the realm of democratic governance, understanding the intricate dynamics of voter preferences is of paramount importance for effective policy formulation & governance. As Abraham Lincoln famously stated, democracy represents a "*government of the people, by the people, for the people*", underscoring the pivotal role of voters in shaping the trajectory of governance through electoral processes. This study embarks on a critical investigation into the factors influencing the electoral decision process of voters within the complex *socioeconomic landscape of India*.

The paper delves into a multivariate analysis of influential factors, ranging from *party affiliation & candidate characteristics* to *policy positions, socioeconomic indicators, social influence, & media impact*. Each of these factors assumes a distinctive role in shaping voter preferences, attitudes, & ultimately, the choices made in the electoral sphere.

By scrutinizing *party affiliation*, we aim to elucidate how individuals' enduring allegiance to political parties influences their voting decisions. A nuanced examination of *candidate characteristics* seeks to discern the impact of personal qualities, qualifications, and leadership attributes on voter perceptions and preferences. This includes analyzing *policy positions* to shed light on how voters evaluate candidates' stances on key issues & the resonance of these positions with their own values & aspirations. Our study delves into the *socioeconomic determinants* of voting behavior, exploring the influence of economic conditions and *demographic characteristics* on voter choices. This involves meticulously analyzing the role of social influence, including the impact of social networks, family, and community. We also investigate the pervasive influence of media on voter perceptions and preferences, considering the implications of media coverage, political advertising, and digital platforms in shaping electoral outcomes. Moreover, we explore the *positive externalities* of the current voting system under the purview of the *Constitution of India*, acknowledging the monumental effort involved in conducting elections for the most populous nation in the world.

Apart from the empirical analysis, this study proposes a ***preference survey of voters*** aimed at garnering *quantitative insights* into the aforementioned factors. Leveraging *Statistical Survey design theory*, the survey will be designed to capture the nuances of voter behavior & preferences, providing valuable data for analysis.

With the Indian Lok Sabha elections of 2024 on the horizon, our research will provide valuable insights into the economic determinants of voter preferences, thereby contributing to the understanding of *electoral dynamics* in the world's largest democracy. It will offer policymakers & political strategists actionable insights for effective governance & democratic representation.

Keywords: Party affiliation, Candidate characteristics, Policy positions, Socioeconomic indicators, Social influence, Media impact, Demographic characteristics, Voter preferences, Positive externalities, Constitution Of India, Statistical survey design theory, Electoral dynamics.

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1 The Problem

Elections are the most important avenue for the citizens of a democratic country to exercise their fundamental right to vote to ensure their participation and influence in the decision-making process in the country. Hence, it becomes imperative for voters to make extremely well-informed choices at the voting booth to ensure that the right candidate comes through who can potentially help the voter's local constituency and, in turn, help the country prosper the most.

The factors on which a voter in a population bases his/her opinion are paramount in determining who becomes victorious in the electoral race. **Through this research, we aim to understand & analyze the parameters based on which voters elect a certain candidate.**

Through our research, we target the use of mathematical methods to uncover hidden patterns and analyze the dependence on the data obtained from our survey on various parameters to determine the potential political leanings of the survey participants. We aim to understand a voter's mentality and how various factors, such as their family background, mode of news consumption, etc., affect their candidate preference and political orientation.

Emphasizing **Fundamental Rights**, **Fundamental Duties**, and **Dharmaarthashastra** in our research highlights the importance of moral and informed voting. This focus aims to guide young Indian voters to consider personal gains, national interests, and ethical standards. By doing so, we nurture a generation of voters committed to the principles that bolster India's democratic fabric and development goals, supporting the country's journey toward becoming a developed nation by 2047 through conscientious and value-driven electoral participation.

2 Motivation

Our motivation to work on this project stems from trying to **gauge voters' preferences**, especially considering the forthcoming **2024 Lok Sabha elections**. Many factors, including socioeconomic status, cultural background, and personal experiences, can influence political orientation, making it a complex and multifaceted phenomenon. Setting up relationships between such complicated factors can be a highly arduous task.

To **perform a rigorous analysis** of the data that we have obtained from our survey. Further, we want to look to explore possible relationships between different variables, which *might help us determine a person's political orientation* and also help us potentially unearth how an individual's responses to questions asked in our survey would indicate their choice of candidate at the voting booth .

3 Objective

Our objective is three folds:

1. Ascertaining the **voting parameters** influencing the contemporary electoral behaviors of the GeneZ students of IIT Kanpur.
2. Finding the **political spectrum distribution** of the sample (Far Left, Center Left, Center, Center Right & Far Right)
3. Exploring the applicability of **Dharmaarthashastra** in the process.

4 Literature Review

The dynamics of voters' preferences have been a topic of study for a long time, and multiple studies have been conducted in this domain in different contexts worldwide.

Our primary inspiration for this research comes from the work of **Kulachai, et. al.**[1], which provides a comprehensive literature review of the factors influencing voting decisions. The authors identify several key factors, including socioeconomic status, political ideology, and candidate characteristics. They also highlight the importance of understanding the context-specific factors influencing voting decisions in different countries. *This paper has largely inspired us to explore the potential relationships between different variables in determining political orientation.* We wanted to perform our analysis to verify the hypothesis put forward by the authors in this article.

We further explored the paper by **Stephanie Tawa Lama-Rewal**[2], which examines the scientific & political debates surrounding election studies in India. The author argues that election studies in India have been shaped by scientific and political considerations, with scholars and politicians seeking to understand the electoral process and its implications for Indian democracy.

Roberto (2017)[3] & **OOSGA (2024)**[4] explored the influence of social media on voting decisions in India, emphasizing the significant role social media plays in shaping public opinion and political discourse, and this study provided us with the motivation to add a question requiring participants to fill in their preferred mode of news consumption. It turns out that most of the Gen-Z population relies on social media to get their news, and hence, this medium becomes of utmost importance in trying to analyze what shaped public opinion in this day and age.

5 On Dataset

To reduce the omitted variable biases and make sure that the sample is sufficiently homogeneous, **the data is extracted from IIT Kanpur Gen Z¹(Those born after 1996) students only** to gather valuable insights and draw potential conclusions on the **political orientation** of each person in our dataset based on certain assumptions & parameters.

Further our study adopts a **gender-agnostic approach**, thereby refraining from introducing gender-based controls.

- **Population:** GenZ IITK Current Students.

There are total 4089 Undergraduates(UGs), 1897 Postgraduates(PGs) & 2250 Doctoral Students ^a. Number of GenZ students will be atleast $(4089 + 1897 + 214) = 6200$

So, **Population Size** ≥ 6200 (This is an approximate calculation to find the lower bound)

^aNIRF 2023

5.1 Survey Design

We designed a survey based on [5] so as to accurately understand the preference set of the target population.

We have conducted two surveys call them **Survey1** & **Survey2** (henceforth : **S1** & **S2**) with two different goals in mind.

The elements out of our sample (ie **Population – Sample**) are similar to the elements inside the sample in all respect. IITK students often tend to have a similar background (same kind of motivations, aspirational class family background and a hunger to learn and achieve more in life; from our practical experiences we have observed them to behave in an often predictably identical way)

All IITK students are connected equally (all of them have smart phones and access to stable highspeed internet connections from the institutes). Provided that we conducted our survey in **google forms** propagated through **WhatsApp** with personal calling and reminders

¹Pew Research Center What We Know About Gen Z So Far

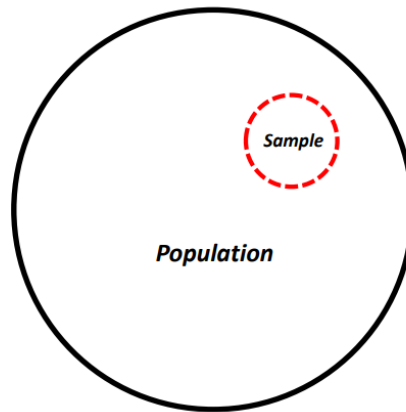


Figure 1: Population vs. Sample

Survey1
• Aim of the S1: Understanding voting preferences of the given population • We consider a sample data: Our sample size is 134. • Mode of Survey: Google Form filled through WhatsApp often by setting personal reminders.
Survey2
• Aim of the S2: Understanding Political Leaning of the population • The sample size has been restricted 134. So as to make it comparable with S1 • Mode of Survey: Same as S1

One important point to note is that we followed a Simple Random Sampling with replacement **SR-SWR**. Therefore a strict inference about the population need not be accurate. **With Replacement** part has been introduced because of the technical constraints of **google form** where if you maintain it anonymous then people can fill the form multiple times.

Questions Asked In the Survey

- In S1, we asked the following questions to our target group to capture:
 1. **Impact of Voter's Family, Candidates' religion & social identity** (like caste, family lineage and other relevant markers)
 - S1.Q1 How much does your family's party affiliation influence your vote?
 - S1.Q6 Does religion of a candidate influence your choice?
 - S1.Q8 Does social identity of the candidate influence your choice?
 2. **Impact of traditional media & traditional political campaigning**
 - S1.Q4 How much do you think the media influence other people's voting preference ?
 - S1.Q7 How much does media influence your political views?
 - S1.Q9 What's your preferred news source?
 - S1.Q12 How much do political campaigns & rallies influence your opinion of candidates and parties?
 3. **Impact on features desired in the candidates by the voters**

S1.Q3 Qualities You Look for in Candidates

S1.Q10 How Important for you is a candidate's track record for delivering on their promise & ensuring accountability?

S1.Q11 How much does a candidate's perceived corruption or ethical conduct influence your vote?

4. Influence of "issues"

S1.Q13 How much does the Local issues influence your vote ? (Considering you are voting in the Lok-Sabha Election)

S1.Q14 How much does the representation of issues related to youth influence your vote ?

5. Checking a particular assertion of [1]

S1.Q2 What do you perceive your personality to be like?

S1.Q5 When voting for a candidate, which of these traits do you prioritise?

6. Participation in the upcoming Loksabha Elections

S1.Q1 Will you vote in the next Lok-Sabha elections?

- In S2 we asked the following 10 questions with only one goal in mind :

1. To decide the degree of Right lean of the voter:

S2.Q1 What do you think your political orientation is?

S2.Q2 Do you think populism should focus on egalitarian ideals?

S2.Q3 Should politics be open to progressive reforms?

S2.Q4 Do you think that the government should be secular in its conduct?

S2.Q5 What is your stance on popular sovereignty being essential for national identity?

S2.Q6 Do you think religion should guide government policies in our democracy?

S2.Q7 Should the government emphasize equality and progress over tradition?

S2.Q8 What is your opinion on the Government's Role in social programs?

S2.Q9 Do you think that the National Identity should be based on shared culture and customs?

S2.Q10 What is your opinion on the government increasing taxes on the wealthy to redistribute wealth?

S2.Q11 What are your thoughts on Economic Policy prioritizing low taxes and less business regulations?

Options for question S2.Q1:

- Center
- Center Left
- Center Right
- Far Left
- Far Right

Options for questions S2.Q2 - S2.Q11:

- Strongly Disagree
- Somewhat Disagree
- Neutral
- Somewhat Agree
- Strongly Agree

A detailed explanations for each questions is provided in the Results section of the paper.

6 Results

6.1 Inferences

In this section we will mention the inferences that we can make from the Dataset of the two surveys that we conducted (S1 & S2) & will simultaneously propose the dynamics/reasons behind observing those inferences, based on the *contemporary political scenario* and the *nature of the cohort* from which the data is collected.

1. Impact of voter's family, candidates' religion & candidates' social identity on the voter:

Here we would refer to responses in **S1.Q1**, **S1.Q6** & **S1.Q8** to make the inferences

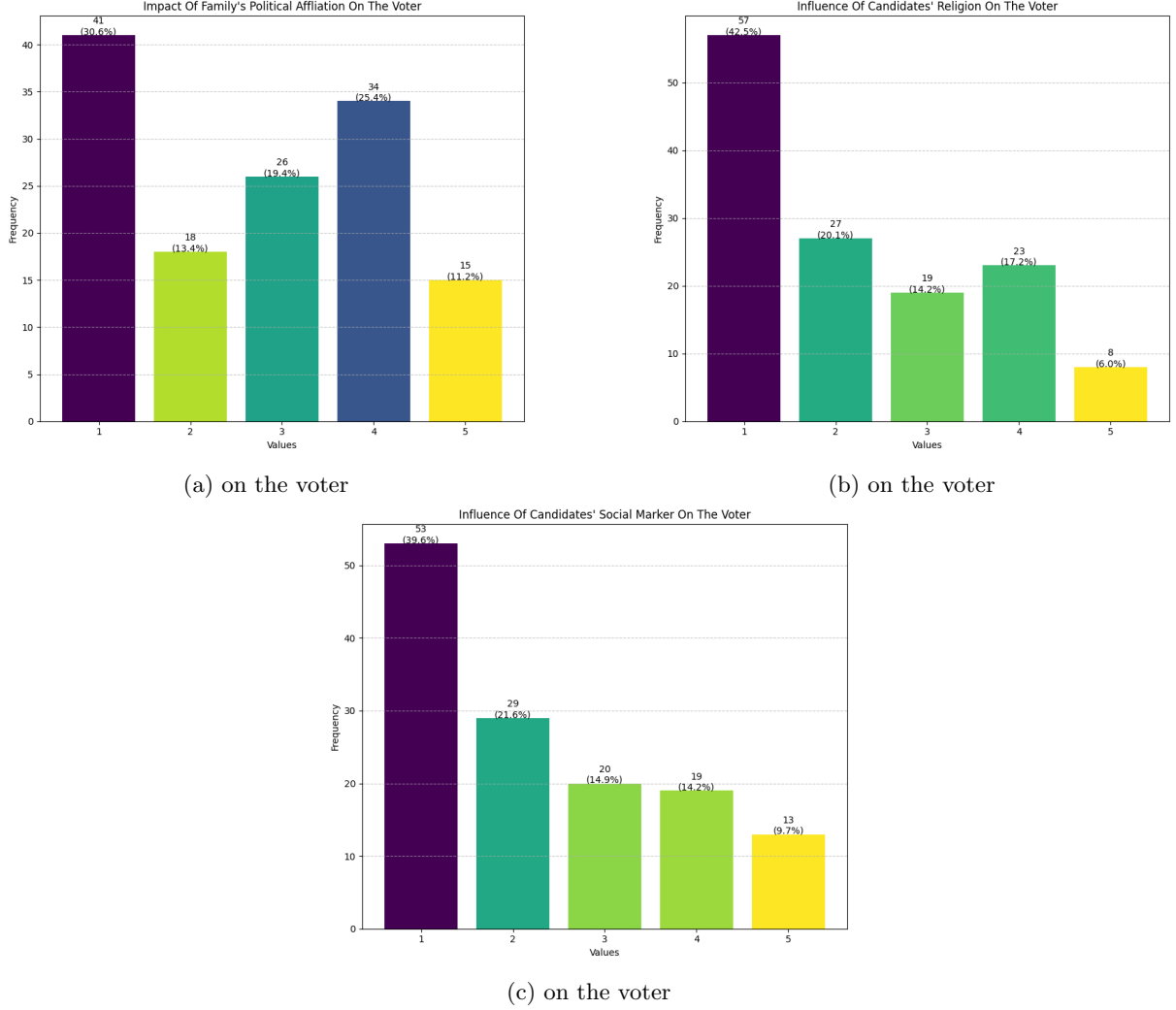


Figure 2: Raw Data: Data collected in the survey

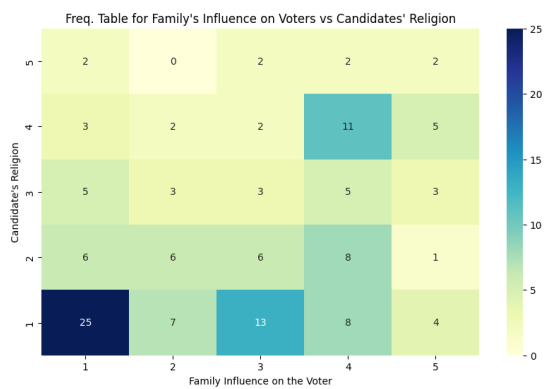
Now we would check the dependence among the above three variables considered in pairs. For this we use χ^2 test to decide whether the variables (considered pair wise) are independent or not ; and if they are not independent then we use the **Cramer's V value** to measure the degree of associations of those variables.

H_0 : The two variables are independent

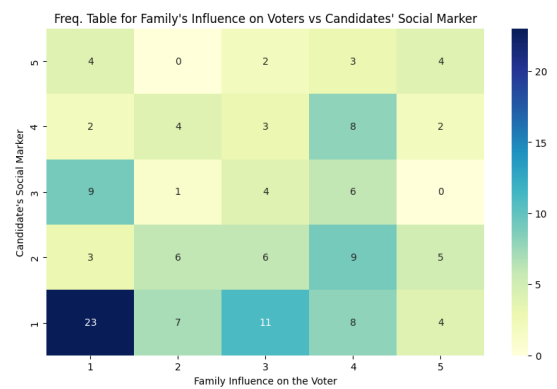
We select a significance level of $\alpha = 0.05$.

If: $p\text{-value} \leq \alpha \implies \text{Reject } H_0 \implies \text{Variable are Dependent}$

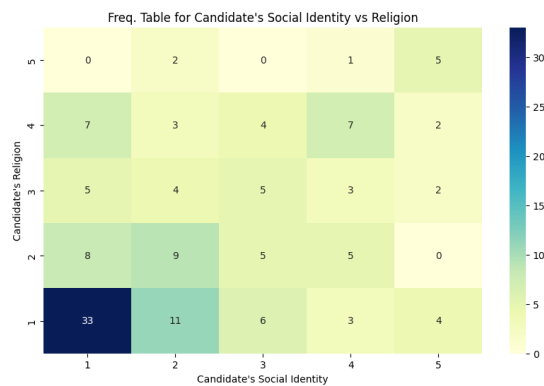
else: Fail to reject $H_0 \implies V$



(a) Family's influence vs. the influence of a candidate's religion on the voter



(b) Family's influence vs. the influence of a candidate's social identity on the voter



(c) Influence of candidate's social identity vs candidate's religion on the voters

Figure 3: Frequency Table: understanding the relationships between the three variables taken two at a time

If we Reject $H_0 \Rightarrow$ There are not enough evidence to suggest that the variables are independent
 $\Rightarrow$ Calculate **Cramer's V** value to evaluate the degree of association between the two variables.

if: **Cramer's V** $\leq 0.1 \Rightarrow$ Small association
 else if: $0.1 < \text{Cramer's V} \leq 0.3 \Rightarrow$ Moderate association
 else $0.3 < \text{Cramer's V} \Rightarrow$ Strong association

Pair of Variables	χ^2 stat	P-value	Hypothesis Testing	Cramer's V
Influence of Family vs Candidate's Religion	25.526	0.061	Fail to Reject H_0 (p-val $> \alpha$)	–
Influence of Family vs Candidate's Social Identity	28.801	0.025	Reject H_0 (p-val $\leq \alpha$)	0.156 (moderate association)
Influence of Candidate's Social Identity vs Candidate's Religion	50.561	0.0	Reject H_0 (p-val $\leq \alpha$)	0.257 (moderate association)

So we can infer that

- Family of the voter has NO impact on voter's preference on candidate's religion:**
 This could be explained by the fact that families from where our cohort belongs are fairly secular in their approach towards electoral candidacy. And family doesnot, ingeneral enforce a strong bias on the voter to elect a candidate from a particular religion.
- Family has a CONSIDERABLE impact on voter's preference towards the candidate's social identity:**
 This result is quite counterintuitive. Even though families don't tend to enforce any religion bias on the candidate on the voter's behalf; but the they **impart quite a strong bias on the voter's preference towards a candidate from a particular social background (caste, familiy lineage and other relvant markers)**.
 It can be explained from the fact that a large respondents from or sample cohort are from Uttar Pradesh, Bihar, Madhya Pradesh, Rajasthan and other such Hindi heartland states; where, historically speaking, there has been an overwhelming presence of **caste based politics**².
Our findings reveal that caste politics continues to exert its influence within our cohort.
- A voter's preference towards religion of the candidate & towards the candidate's social identity is SIGNIFICANTLY correlated**
 Its very natural for those with strong religious bias to have an explainable preference for social markers of the candidates . Therefore this result is as per our expectations.

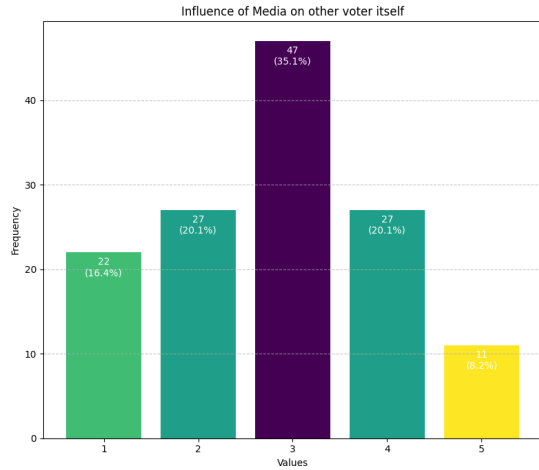
2. Impact of Media and traditional political campaigning

To make relevant inference we refer to response of questions **S1.Q4, S1.Q7, S1.Q9 & S1.Q12**
 we conduct χ^2 test to check the dependence of these two variables

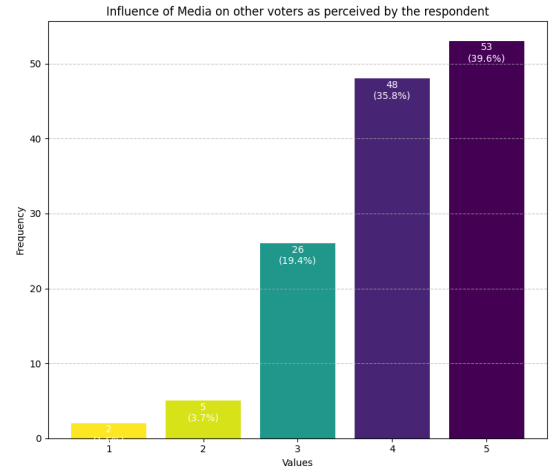
Pair of Variables	χ^2 stat	P-value	Hypothesis Testing	Cramer's V
Influence of Media on the Other Voters vs the Voter itself	41.627	0.0	Reject H_0 (p-val $\leq \alpha$)	0.222 (moderate association)

- There is a SIGNIFICANT association between the influence of media on the voter & the perceived influence on other voters:**
 As is visible from the plot5a & 5b people tend to underweigh their own media influence and expect the media to have a strong impact on other population, which is quite unique.

²wiki: Caste Politics In India

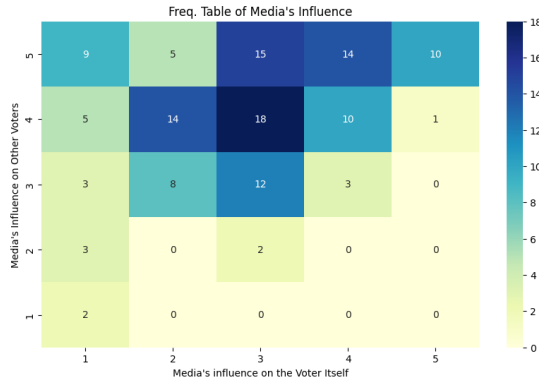


(a) Media Influence on the respondent itself

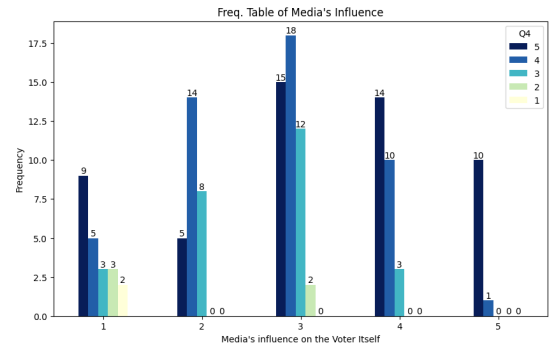


(b) Perceived media influence on other voters

Figure 4: Raw Data: Media's influence on the respondent vs the influence on other voters as perceived by the respondent



(a) Frequency plot 1



(b) Frequency plot 2

Figure 5: Frequency Plot: Media's influence on the respondent vs the influence on other voters as perceived by the respondent

(b) **Increased consumption of Internet based Media:**

From 6 we infer that an overwhelming number (67.9%) of folks in our cohort consumes **Social media** (exp. Facebook, Twitter, Youtube). Followed by **Newspapers & News websites**.

Only 32.8% refers to regular **TV based news channels** when consuming news. This suggest an increased appetite for internet based media among GenZ; which is quite expected as GenZs are the first truly **Digital Natives**

(c) **Almost no impact of "Traditional" political campaigning:**

From figure 14 we can decipher that $\approx 90\%$ of our cohort had moderate to NO effect of "traditional campaigning". By traditional campaigning we mean political rallies, speeches & processions.

3. **Impact on Qualities desired in candidates by the Voters:**

We refer to questions **S1.Q3**, **S1.Q10**, **S1.11** and **S1.Q5** & **S1.Q2**

From the figures 8, 9a & 9b we can make the following inferences:

(a) **Integrity & honesty, Ability to lead, Future vision** are the top 3 qualities that are being preferred by our cohort. While **Ability to work across the party lines** is the least preferred quality. This result is in line with [1] .

A point to note is that $\approx 40\%$ of the respondents do care about **environment**.

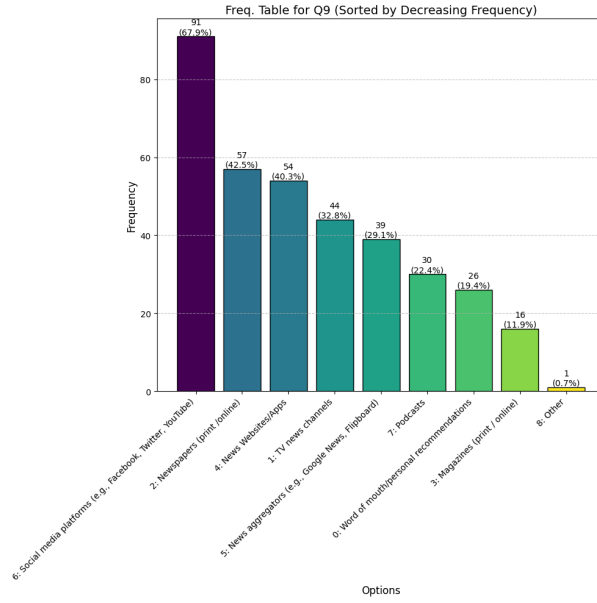


Figure 6: Preferred media of the respondent (voter)

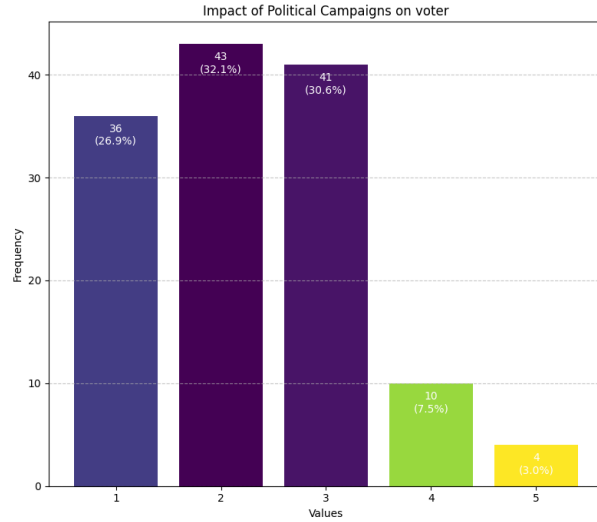


Figure 7: Impact on Political Campaign on the voter

From this we can argue that our cohort seeks a candidate with honesty, strong leadership, future vision, transparency, accountability, and good academic background.

- (b) We have observed that **Integrity & honesty** are strongly preferred by the electorate. **S1.Q11** answers the degree upto which people prefers **Candidate's ethics**.

As one can observe from fig. 9b that a whopping $\approx 77\%$ respondents gives a score of $4 \rightarrow 5$; thereby underscoring the importance of ethics in the candidates.

- (c) One can observe in 8 that 63.4% prefers **Transparency & Accountability** and 49.3% finds **Track records & experience** to be important.

S1.Q10 intended to measure the same. From 9a we found out that $\approx 80\%$ rated it $4 \rightarrow 5$

4. Checking an assertion of Kulachai, et. al [1]

The paper[1] suggested that an **"Introvert"** person prefers **"Reserved & Thoughtful"** candidate and an **"Extrovert"** person prefers **"Charismatic & Assertive"** candidate.

To address this particular question we use the response of **S1.Q2 & S1.Q5**

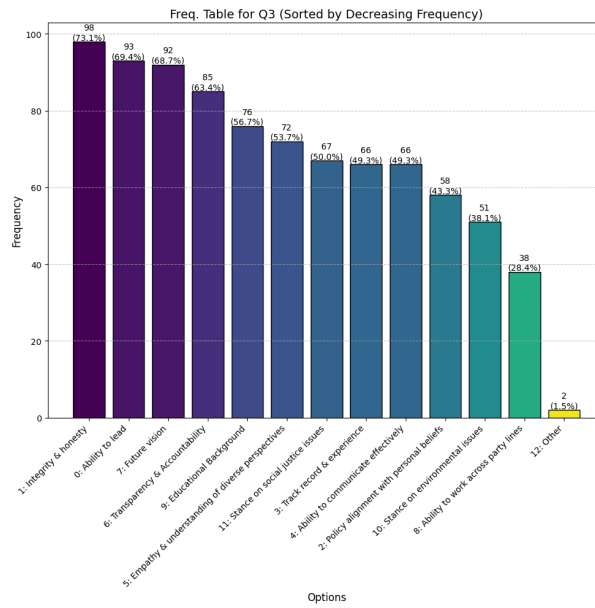
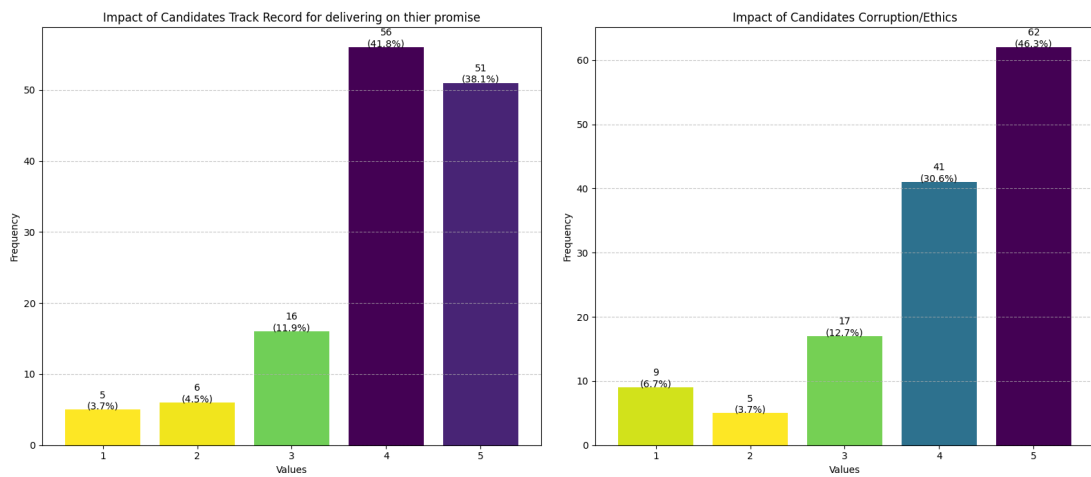


Figure 8: **Qualities in Candidates** that are preferred by the voters



(a) Influence of **Candidates' track record on delivery & accountability** on the voter (b) Influence of **candidate's Ethical conduct & ccorruptions**

Figure 9: Raw Data: As collected in the Survey for these three variables

We have mapped voter's personality: **Introvert**→ 0, **Extrovert**→ 1 & **Ambivert**→ 2 and Candidates' nature: **Charismatic & Assertive** → 0 and **Reserved & Thoughtful** → 1

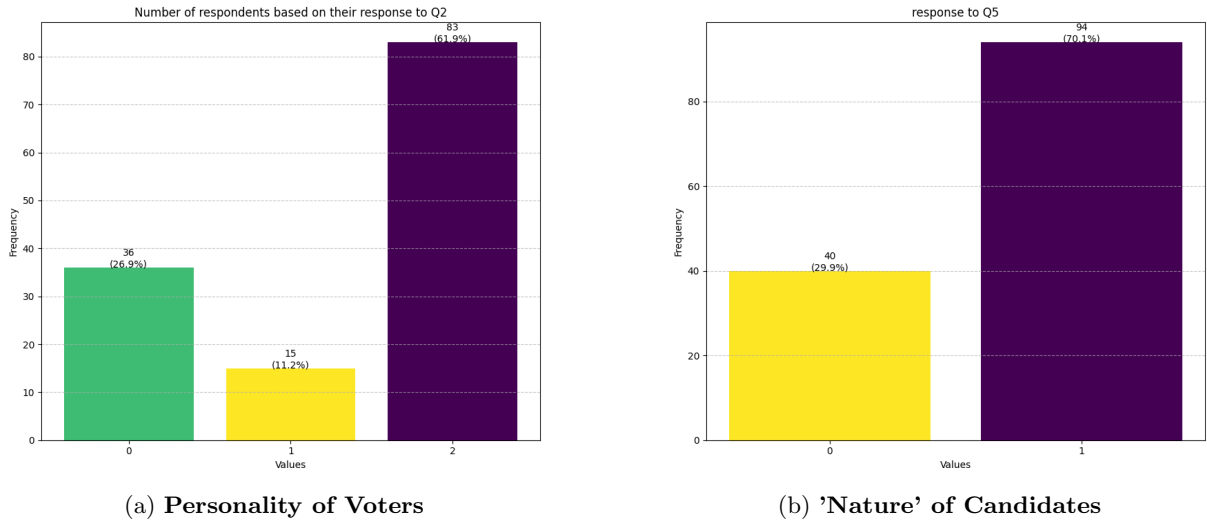


Figure 10: Raw Data: **Voter's Personality** and the **Candidates' Nature**

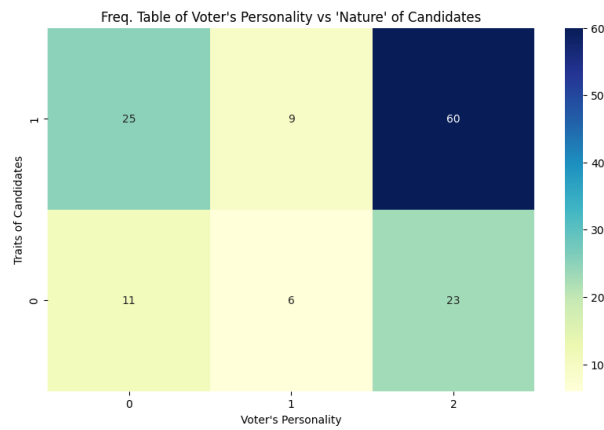


Figure 11: **Voter's Personality** vs **Candidates' Nature**

From the fig. 10 and 11 we infer the following:

$\frac{n(\text{"Reserved \& Thoughtful"})}{n(\text{"Charisma \& Assertive"})}$ Personality of Voters	vs	Introvert	Extrovert	Ambivert
		2.273	1.5	2.609

Table 1: Calculated ratios

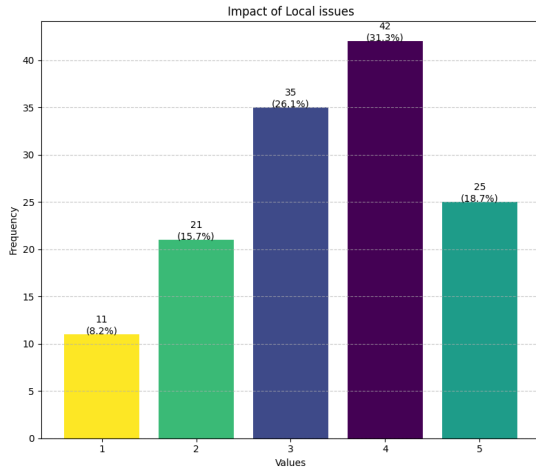
5. Impact of particular type of issues

We measured the impact of "**Local issues**" in a national level election (Lok Sabha election in India). Further we also measured how the **issues related to youth** impact our cohort's voting preference.

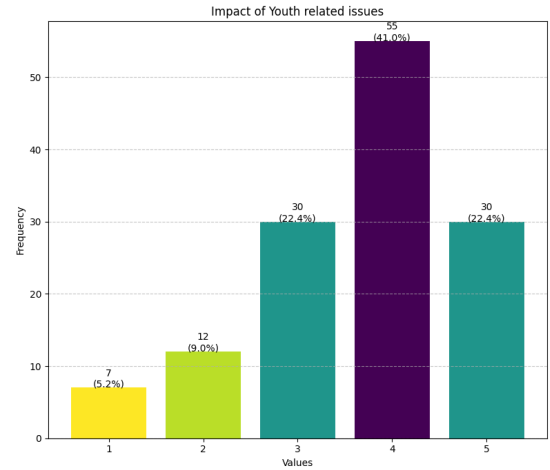
By "local issues" we meant issues pertaining to a particular constituency such as building a hospital, providing proper public services & amenities, providing of local schooling etc.

By "Youth related issues" we meant political empowerment of youths and meeting their aspirations (proper job market, transparency, higher education facilities etc).

For this we refer to responses of **S1.Q13** & **S1.Q14**



(a) Impact of "Local Issues"



(b) Impact of "Youth Related Issues"

Figure 12: Impact of Various issues

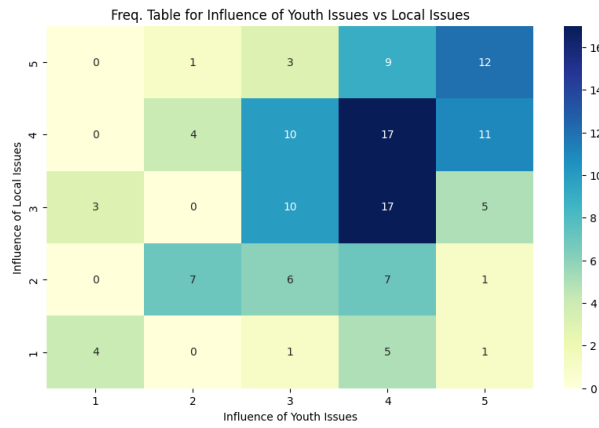


Figure 13: Impact of "Youth Issues" vs "Local Issues"

From fig. 13 & 14 we can infer the following:

(a) **Local Issues do matter in a national level election**

This may sound quite contrarian; but even among our folks (where a good number of them come from a economic middle class background) we observe a large priority for "Local issues" while voting.

(b) From figure 12b, as expected, we infer that issues related youth do worries a lot of GenZ. $\approx 80\%$ folks have rated it from 3 $\rightarrow$ 5

6. **Expected participation in the upcoming Lok Sabaha 2024 election:**

$\approx 80\%$ of the respondents are intending to vote this Lok-Sabha election 2024. Refer fig. 14

7. **Ideology Preference of Individual (we will be analyzing S2 responses)**

Exploring ideologies using **left-right leaning** offers an understanding of the diverse political landscape globally. The ideas of left-leaning and right-leaning ideologies come from the **French Revolution**, which encapsulates the fundamental ideology divide between hierarchy and social equality. There is a segregation of ideology between **left-wing** and **right-wing**, where **left-wing** properties are briefly defined as emphasizing more **social equity** and **government intervention**. Conversely, the **right wing** prioritizes **individual liberty** and **economic freedom**, providing a framework for analyzing political behavior and party systems.

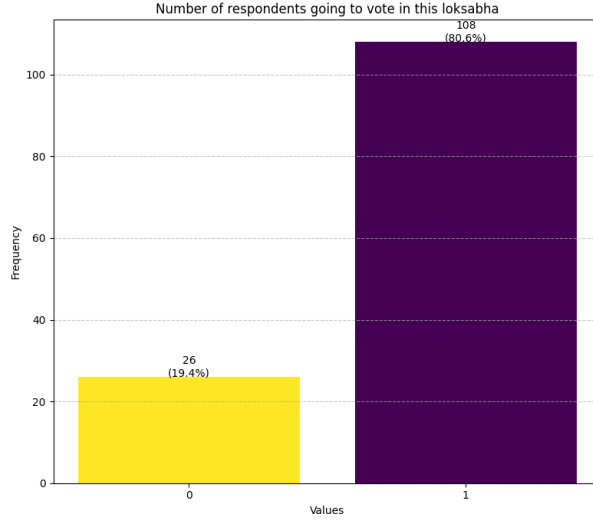


Figure 14: Number of respondents who are going to vote

Considering the case of India, as we all know, the BJP(Bhartiya Janta Party) and the Indian National Congress are the modern embodiments of these ideological divides. The traits of the BJP align more with right-wing principles, mainly the focus of the BJP on the Hindu religion, and the strong central government has resonated with a large portion of the Indian electorate, as evidenced by its electoral success over the past decades. Opposite to the BJP, the Congress party’s left-leaning ideology centered around secularism and social welfare has seen declining support from the 2014 general election.

We have floated a survey form S2 among the IITK GenZ students to capture the political spectrum distribution³. The study’s classified respondents into five distinct political orientations.

- (a) Far Right
- (b) Center Right
- (c) Center
- (d) Center-Left
- (e) Far left

The classification is done based on responses to a set of questions in S2. The explanation concerns the right wing—only the left-wing exhibits opposite characteristics. **Classification of Political Orientation Based on Responses to Survey Section S2**

This summary outlines the classification of political orientation based on individual responses to various statements regarding government role, economic policy, national identity, and other socio-political beliefs:

- (a) **Government Role in Redistribution:**
 - *Disagree*: Right-wing (Prefers lower taxes, less redistribution)
 - *Agree*: Left-wing (Supports increased taxes on the wealthy for redistribution)
- (b) **Economic Policy Preferences:**
 - *Agree*: Right-wing (Advocates for low taxes and minimal business regulation)
 - *Disagree*: Left-wing (Favors more regulatory oversight and tax policies targeting inequality)
- (c) **Expanded Government Role in Social Programs:**
 - *Disagree*: Right-wing (Supports limited government involvement)
 - *Agree*: Left-wing (Advocates for expanded government role)

³Inspired from *Participatory ideology: From exclusion to involvement*; Peter Beresford

- (d) **National Identity Based on Shared Culture and Customs:**
 - *Agree*: Right-wing (Values a unified national identity based on tradition)
 - *Disagree*: Left-wing (Supports a more inclusive or diverse national identity)
- (e) **Influence of Religion on Government:**
 - *Somewhat agree*: Right-wing (More open to religious influence on policies)
 - *Disagree/Somewhat disagree*: Left-wing (Prefers separation of church and state)
- (f) **Emphasis on Equality and Progress Over Tradition:**
 - *Disagree*: Right-wing (Prioritizes tradition over progressive changes)
 - *Agree*: Left-wing (Focuses on equality and progressive reforms)
- (g) **Secular Governance:**
 - *Disagree*: Right-wing (Less emphasis on strict separation of religion and state)
 - *Agree*: Left-wing (Strong advocate for secular governance)
- (h) **Popular Sovereignty as Essential for National Identity:**
 - *Agree*: Right-wing (Emphasizes national sovereignty)
 - *Disagree*: Left-wing (May prioritize global or communal approaches over national sovereignty)
- (i) **Focus of Populism:**
 - *Neutral*: Right-wing (Focuses on national sovereignty and traditional values)
 - *Strongly agree*: Left-wing (Emphasizes empowerment and socio-economic reforms)
- (j) **Openness to Progressive Reforms:**
 - *Disagree*: Right-wing (Cautious about progressive changes)
 - *Agree*: Left-wing (Supports comprehensive progressive reforms)

- **Applying some ML models to predict the political Orientation:**

Machine learning is slowly changing various industries and Aspects of daily life worldwide. Machine learning is everywhere, including healthcare, finance, retail, manufacturing, Agriculture, Entertainment, security, customer service, etc. Data science and machine learning are slowly replacing humans in almost every aspect of life. Analysts in the political landscape also utilize machine learning by classifying individual preferences into broader political spectrums. This classification aids political entities in understanding constituency dynamics and formulating targeted campaigns. To investigate the efficacy of machine learning in this area, we conducted a survey incorporating ten carefully curated questions that implicitly categorize individuals as 'far-left,' 'center-left,' 'center,' 'center-right,' and 'far-right.' Lastly, we asked them about their ideology for getting a supervised dataset.

The survey yielded a labeled dataset essential for supervised machine-learning techniques. We implemented four primary classification algorithms: **Logistic Regression**, **Decision Tree**, **Random Forest**, and **Support Vector Machine (SVM)**. The performance of these classifiers was evaluated based on their prediction accuracy for the political leaning of individuals.

The **Logistic Regression** model demonstrated a prediction accuracy of **74%**. The **Decision Tree** classifier outperformed **Logistic Regression** with an accuracy of **81%**. This increase could be attributed to the Decision Tree's capability to map non-linear relationships common in opinion-based data. Interestingly, the **Random Forest** classifier, a more robust version of Decision Trees through ensemble learning, did not show an improvement and matched the Logistic Regression accuracy at **74**. The SVM's performance was similar to the aforementioned models.

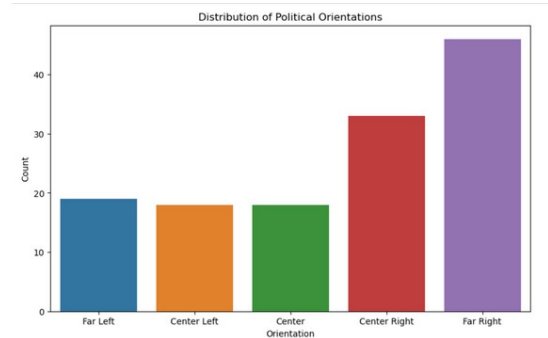
One critical observation was that the dataset comprised only **134** responses, a relatively small sample size that may hinder the model's ability to generalize; this is evident in the moderate accuracy scores and potential **underfitting** issues. A small dataset limits the complexity of the patterns that the model can learn, hence potentially failing to capture the general trend of the population's political leanings.

We suggest two potential methods to mitigate this limitation: **data augmentation** and **oversampling**. Data augmentation artificially increases the size of the dataset by generating

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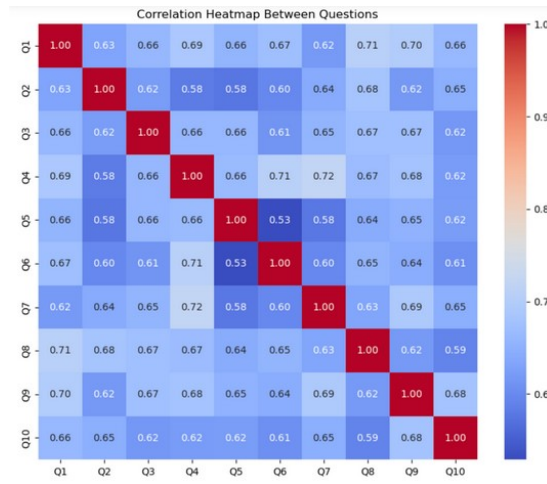
Orientation
Center      18
Center Left 18
Center Right 33
Far Left    19
Far Right   46
dtype: int64

```



(a) showing proportion of Different ideologies in our Survey

(b)



(c) Correlation Heatmap

Figure 15: Depiction of number of responses for each political orientation type and their correlations

new samples from existing ones, while oversampling involves replicating samples from the minority class to balance the dataset. Both techniques could enhance the model's learning capacity, enabling it to generalize better to unseen data.

Integrating machine learning to predict individual preferences can significantly benefit political parties and analysts. It can streamline understanding demographic trends, refine policy communication, and strategize electoral approaches tailored to diverse voter segments. Though constrained by data volume, the preliminary results lay a promising foundation for how machine learning could revolutionize data-driven political strategy in the digital era.

- **How Our Model works**

This short Paragraph describes how our model predicts new individuals' preferences. This time, we asked individuals only those 10 survey questions and not his/her ideology preference since we already trained our model using label data. The response filled by new individuals acts as test data we call predict function with the obtained response and our model will give ideology preference of an individual.

6.2 Study specific suggestions

The first page of our constitution, The Preamble, lays down the core values and ethics that a citizen of our country should abide by in their conduct, representing the nation's ideals. These principles are meant to guide us daily, especially when we vote. It's about choosing leaders who embody these principles and ensuring our decisions at the ballot box mirror the high standards our nation strives for. One of the keywords in our preamble is that we are a "Secular" republic, and hence, definitely, our voting decisions should not be driven by the religion of the candidate. But as we have observed from our survey, a large chunk of responders voted in the affirmative of a candidate's religion being a major factor while deciding

Accuracy: 0.7407407407407407					Accuracy: 0.8148148148148148				
Classification Report:					Classification Report:				
	precision	recall	f1-score	support		precision	recall	f1-score	support
Center	0.00	0.00	0.00	1	Center	0.80	1.00	0.89	4
Center Left	0.50	1.00	0.67	4	Center Left	0.50	1.00	0.67	2
Center Right	0.70	1.00	0.82	7	Center Right	1.00	0.62	0.77	8
Far Left	1.00	0.33	0.50	6	Far Left	1.00	0.33	0.50	3
Far Right	1.00	0.78	0.88	9	Far Right	0.83	1.00	0.91	10
accuracy			0.74	27	accuracy			0.81	27
macro avg	0.64	0.62	0.57	27	macro avg	0.83	0.79	0.75	27
weighted avg	0.81	0.74	0.72	27	weighted avg	0.87	0.81	0.80	27

(a) Logistic Regression Accuracy					(b) Decision Tree Accuracy				
Accuracy: 0.7407407407407407					Model Accuracy: 70.37%				
Classification Report:					Classification Report:				
	precision	recall	f1-score	support		precision	recall	f1-score	support
Center	0.00	0.00	0.00	1	Center	0.00	0.00	0.00	1
Center Left	0.57	1.00	0.73	4	Center Left	0.50	1.00	0.67	4
Center Right	0.64	1.00	0.78	7	Center Right	0.64	1.00	0.78	7
Far Left	1.00	0.50	0.67	6	Far Left	1.00	0.33	0.50	6
Far Right	1.00	0.67	0.80	9	Far Right	1.00	0.67	0.80	9
accuracy			0.74	27	accuracy			0.70	27
macro avg	0.64	0.63	0.59	27	macro avg	0.63	0.60	0.55	27
weighted avg	0.81	0.74	0.72	27	weighted avg	0.79	0.70	0.68	27

(c) Random Forest Accuracy					(d) Support Vector Machine Accuracy				
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Figure 16: The accuracy of machine learning models in predicting an individual’s preferences.

to hand over the reins of their constituency for the next 5 years. This is where the self-consciousness or the **Dharmaarthashastra** of each individual towards our great nation should become paramount so that they are ready to perform their Fundamental Duties to the fullest, and if we follow our Dharma only then can we think about achieving our goals of becoming a developed nation by 2047.

We have also explored how family affiliations to a political party influence the decision-making of the youth when choosing which candidate to vote for. Many of our survey participants are highly influenced by the choices of their family members while voting. This should not be the case. A family should always educate their children or young adults about how they need to make informed choices rather than blindly follow a family’s affiliation. This is highly imperative because, as seen from our survey, for different individuals, different qualities in candidates are more important than the others.

Suppose a person at their core values is a right-leaning person, but their family has historically voted for the left-leaning parties. If the person blindly follows their family’s footsteps, then their actual characteristics won’t be reflected in the person that comes to power, and this could lead to the weakening of the opposition as well, which can never be a good thing for democracy in the long run.

Our survey gathered that most people in the survey demographic rely on social media to get their daily news. With every other medium, this mode of communication offers several caveats regarding the authenticity of the information. It is therefore necessary for users to be mindful and should perform thorough research and fact-checking of whether the news that they have been exposed to is legitimate or not in the first place before forming their opinions on certain issues and, in the case of elections, the candidates.

7 Summary & Conclusion

In this study, we investigated the factors that influence the choices made by Gen Z voters in India, specifically focusing on how their political affiliations, the characteristics of candidates, the socio-economic environment, and the media landscape affect their preferences. We used our surveys to try and disentangle the intricate web of influences that shape the voting patterns and opinions of this important group of young people. Our results provide a window into the complex and multifaceted realm of Indian electoral behavior.

From our research, while trying to find out relationships between different variables, we can conclude the following:

1. **Family Influence vs. Social Identity and Religion:** According to our study, family influence significantly impacts voters' preferences towards candidates' social identity, such as caste and family lineage, especially in regions with a strong tradition of caste-based politics. However, family influence does not significantly affect the voter's preference based on the candidate's religion, indicating a more secular approach in electoral decisions within the surveyed population.
2. **Media's Role:** The results show that Gen Z voters increasingly consume internet-based media, with social media as their primary source of news and information. Traditional media and political campaigning have less impact, showing the growing importance of digital platforms in molding political ideas and decisions.
3. **Candidate Characteristics:** Voters prioritize three traits in candidates: honesty, integrity, and leadership abilities. This choice highlights the public's need for accountable and honest leadership. Furthermore, a sizable percentage of respondents highly emphasize a candidate's moral character and record, demonstrating the need for integrity in politics.
4. **Importance of Local and Youth-Related Issues:** Voting decisions are heavily influenced by local issues and youth-related concerns, indicating that voters are interested in addressing issues that impact the younger population and urgent and community-specific requirements. This focus on topics pertaining to youth and the local community represents a more practical and forward-thinking approach to voting among Generation Z voters.
5. **Voting Participation and Political Orientation:** Strong political participation among the group polled is indicated by the high intent to vote in the future elections. In addition, the study looks at the political spectrum from left to right, indicating that the Gen Z population has a wide range of ideologies.

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